ME 264 Mechanisms Synthesis & Analysis I
Unit 1 Fundamentals: Terminologies and Definitions

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RECOMMENDED TEXTBOOKS FOR FURTHER READING

1. Robert L. Norton, Design of Machinery An introduction to Synthesis and Analysis of Mechanisms and Machines, 2nd Edition, McGraw-Hill. 1999
2. C. E. Wilson and J. P. Sadler, Kinematics and Dynamics of Machinery, Prentice Hall, 2003.
3. A. G. Erdman and G. N. Sandor , Mechanism Design, Analysis and Synthesis, Vol. I 2nd Edition, Prentice Hall, 1991.
4. Joseph E. Shigley and John J. Uicker, Jr, Theory of Machines and Mechanisms, McGraw-Hill, 1981.
5. Charles E. Wilson and J. Peter Sadler, Kinematics and Dynamics of Machinery, Harper Collins College Publishers.
6. J. Hannah & R. C. Stephens, Mechanics of machines -Elementary theory and Examples, Edward Arnold.
7. R. S. Khurmi & J. K. Gupta, Theory of Machines, Eurasia Publishing House (PVT.) Ltd, 2005.

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COURSE OUTLINE

1. Mechanisms-Definitions & terminology
2. Formalized Dimensional Graphical Synthesis of Linkage
   (a)Two Position synthesis without quick-return
   (b)Addition of Drive dyad and use of Drive cylinders
   (c)Two Position synthesis with quick-return
   (d)Three Position synthesis with and without specified pivots

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COURSE OUTLINE

3. Displacement and Velocity Analysis of planer mechanisms (up to six-bar linkage)
   - Analytical method using vectors
   - Graphical method
     - Instantaneous centre method
     - Relative velocity method
4. Acceleration Analysis (Vector analytical method only)
5. Transmission of Rotational Motion
   - Gear trains including planetary gear (epicyclic) trains
   - Belt and rope drives

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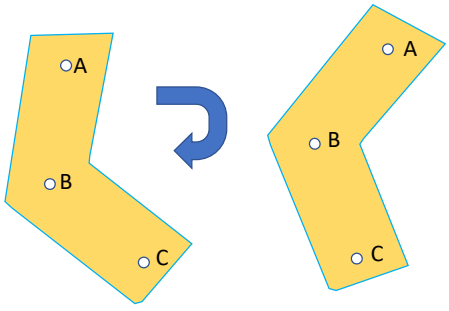
FUNDAMENTALS OF MECHANISMS

- Mechanisms are basic units of machines. Mechanisms are all variations of linkages. Sample machines include cranes: wrapping machines block-making machines, car jacks, bottling machines, agricultural machines, optical drives, sewing machines, timing mechanisms, standing fans, etc.
- Machines consist of simple to complicated mechanisms. In practice, selecting a mechanism to perform a function in a machine is very challenging.
- Mechanism design and analysis require experience and good background in mathematics and dynamics.
- Kinematics is the study of the relationship between time, position, velocity and the acceleration of interacting rigid bodies.

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Terminologies and Definitions

- ***Rigid body***
  - A ***rigid body*** is an idealised model of an object that does not deform or change in shape.
  - Thus, the distance between any two points on a rigid body remains constant under the action of forces.
  - Machine components that do not fit this assumption such as springs in either tension or compression and, belt or rope under compressive load are not links.



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Terminologies and Definitions: Kinematic Link or Element

- It is a part of machine, which moves relative to some other part.
- A link may consist of several parts, which are rigidly fastened together, and hence there is no relative motion between each pair of parts of the link.
- A link or element needs not to be rigid body, but it must be capable of transmitting the required force with negligible deformation.

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Terminologies and Definitions: Kinematic Link or Element

- The main characteristics of a link are:
- There is relative motion between itself and other links
- It should be capable of transmitting a force, torque and/or motion with little or no deformation.

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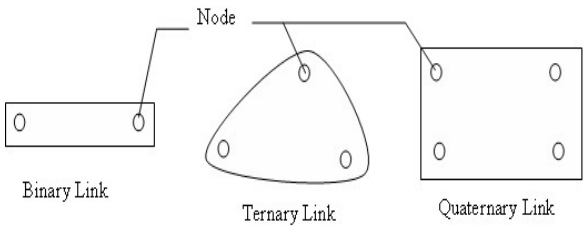
Terminologies and Definitions: Linkage

- A rigid body of any shape that has some number of attachment points, called nodes.
- Linkages are connected specifically to have a controlled output.

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Types of Links: Rigid Link

- A rigid link is capable of transmitting motion, force and/or torque with no or negligible deformations.
- It possesses at least two nodes, which are points for attachment to other links.



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Types of Links: Flexible Link

- It is capable of transmitting motion, force and/or torque with deformation that does not affect its performance.
- Examples of flexible links include belts, ropes, chains and wires. However, belts, chains and ropes have one-way rigidity.
- Therefore, they are considered as links when they are in tension and non-link when they are in compression.

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Types of Links: Fluid Links

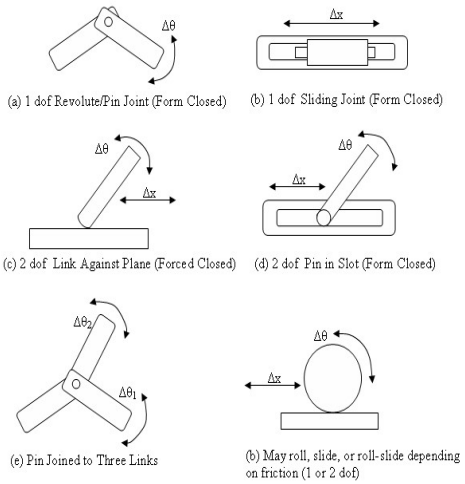
- It is formed by a fluid in a receptacle. In this case, motion and force or torque are transmitted through the fluid pressure.
- Examples include Hydraulic and pneumatic devices

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Joint or Kinematic pair

- It is connection between two or more links, which permit some or potential motion between the links.

- Joints may be classified by:
  1. the type of contact between the links, e.g. point, line or surface.
  2. the number of degree of freedom allowed by the joint.
  3. the type of physical closure at the joint: The closure be either force or form closed.
  4. the number of links joined together.



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- Kinematic Chain
  - It is an assembly of links and joints, which are interconnected such that it has internal mobility and able to transform a supplied motion into a controlled output.

- Mechanism
  - It is a kinematic chain with at least one link grounded or fixed to a reference frame. Three general tasks for mechanisms include the generation of path, motion and function.

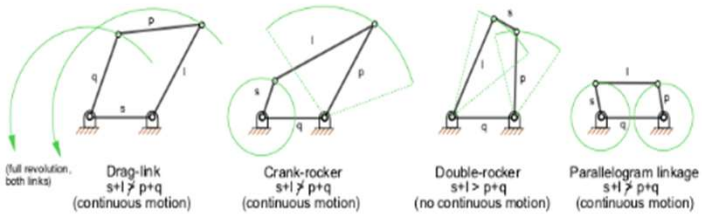
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- Machine
  - It is an assembly of rigid bodies such that it has internal mobility and capability to apply power, transform motion or/and do work.
  - The principal aims of machines are to produce a desired motion, force, torque, work or power.

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Kinematic Inversion

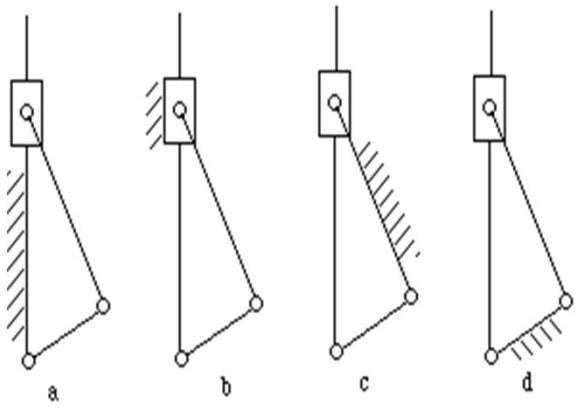
- The absolute motion of a linkage depends on which link is fixed.
- An inversion is created by grounding a different link in a linkage.
- If two equivalent linkages have different fixed links, then each linkage is an *inversion* of the other.



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Kinematic Inversion

Inversions created by grounding different links in a crank-slider linkage



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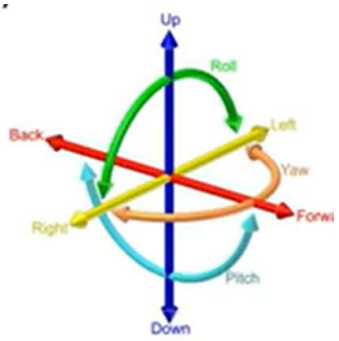
Differences between Machine and Structure

Machine	Structure
Relative motion between parts	No relative motion between parts
Energy is transformed into useful work	No energy is transformed into useful work
Members of a machine may transmit both power (or force, torque) and motion	Members of a structure may transmit only force and moments

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Degree of Freedom

- It is defined as the number of independent parameters that define its configuration.
- It is the number of parameters that determine the state of a physical system.



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Degree of Freedom (or Mobility) in Planar Mechanisms

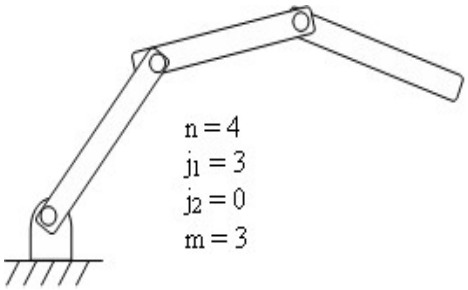
It is defined as the number of independent inputs required in order to predict the output of a mechanism is called *mobility* or *degree of freedom*. The degree of freedom (dof) or mobility of a mechanism with one ground link is given by

$$m = 3(n-1) - 2j_1 - j_2$$

where n = number of links, j_1 = number of 1 dof joints, and j_2 = number of 2 dof joints. The above equation is Kutzbach criterion or modified Gruebler's equation.

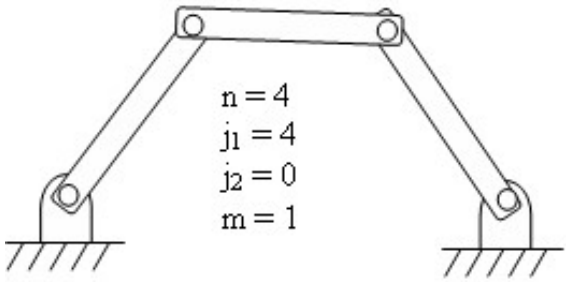
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Examples



$$m = 3(n-1) - 2j_1 - j_2$$

m=3(4-1)-2(3)-0=3

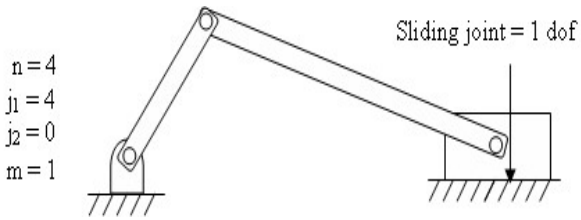


$$m = 3(n-1) - 2j_1 - j_2$$

m=3(4-1)-2(4)-0=1

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Examples

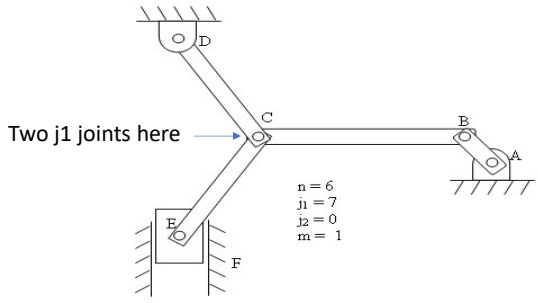


$$m = 3(n-1) - 2j_1 - j_2$$

m=3(4-1)-2(4)-0=1

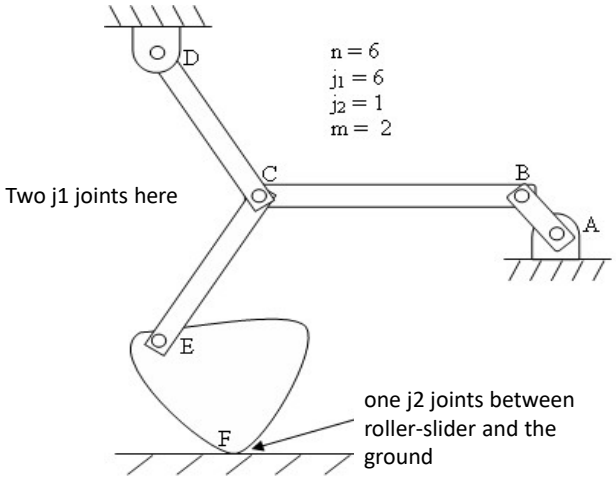
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Example



$$m = 3(n-1) - 2j_1 - j_2$$

$$m=3(6-1)-2(7)-0=1$$



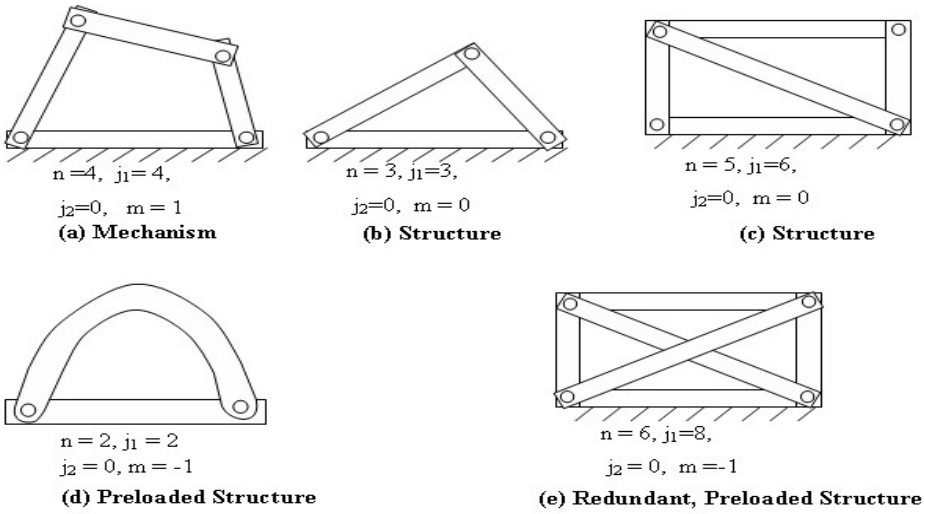
$$m = 3(n-1) - 2j_1 - j_2$$

$$m=3(6-1)-2(6)-1=2$$

Mechanisms and Structures

- Degree of freedom (dof) of an assembly predicts whether an assembly is a mechanism or a structure.
- If dof is positive (i.e. m ≥ 1), then the assembly is a mechanism and there is relative motion between its links.
- If dof is zero (i.e. m = 0), then the assembly is a structure and no motion is possible.
- If dof is negative (i.e. m ≤ -1), then the assembly is preloaded or redundant and no motion is possible.
  - Preloaded structures refer to structures having stresses at the time of assembly with any external load applied.
  - Redundant structures are statically indeterminate, and have redundant link(s).

Mechanisms and Structures



Inconsistencies of Gruebler's Equation

- Sometimes, the Gruebler's equation gives false mobility. For example, Figures (a), (b) and (c) have same calculated DOF value 0 using the Gruebler's equation.
- However, Figure (a) and (c) move and Figure (b) does not move.
- The difference in behaviour with same dof is due to omission of dimensions of links in the Gruebler's equation.
- Inclusion of dimensions would have made the equation lengthy and complicated, thus making it less useful for dof calculation.

